

Functional Principal Components-Based Multiple Imputation for Functional Data

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Abstract Text:

Missing data are a common problem in biomedical research. Valid approaches for dealing with missing data have been proposed and are regularly implemented in applications where the data are exclusively scalar-valued. However, with advances in technology and data storage, biomedical studies are collecting functional data with increasing regularity – and these functional data may be subject to missingness. We propose extensions of multiple imputation via functional principal components regression for missing functional data. This approach is evaluated via a simulation study and applied to data from a study of subjects with major depressive disorder for which both clinical (scalar) and imaging (functional) data are available.

Keywords:

functional data analysis|missing data|multiple imputation|functional principal components|

Can this be considered for alternate subtype?

No

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